Lecture 2: Software Modeling and Analysis

Module 1 | Computer Architecture / Software Architecture

IIIT Nagpur

What is Software Modeling?

- Software modeling is the process of creating abstract representations of a software system.
- A model helps us understand, analyze, and communicate system behavior and structure before implementation.
- Models reduce complexity by focusing only on relevant aspects of the system.

Why Software Modeling is Needed

- Real-world software systems are complex and difficult to understand as a whole.
- Modeling helps identify errors, inconsistencies, and missing requirements early.
- Early modeling reduces development cost and risk.

Model vs Real System

- A model is a simplified version of the real system.
- It does not include every detail, only what is necessary for understanding or analysis.
- Just like a map represents a city, a model represents a software system.

Abstraction in Software Engineering

- Abstraction is the technique of hiding unnecessary details.
- It allows developers to focus on what a system does rather than how it is implemented.
- Abstraction is fundamental to modeling and architecture.

Types of Abstraction

- Data abstraction focuses on what data is stored.
- Functional abstraction focuses on what operations are performed.
- Architectural abstraction focuses on system structure.

Software Analysis – Definition

- Software analysis is the activity of understanding and defining what the system should do.
- It focuses on the problem domain rather than the solution.
- Analysis is performed before design and implementation.

Objectives of Software Analysis

- Understand user requirements clearly.
- Identify system boundaries and constraints.
- Ensure correctness and completeness of requirements.

Analysis vs Design

- Analysis describes what the system should do.
- Design describes how the system will do it.
- Analysis is problem-oriented, design is solution-oriented.

Analysis Modeling

- Analysis modeling represents system requirements using diagrams and models.
- It helps visualize system behavior and data flow.
- Models serve as communication tools between stakeholders.

Best Practices in Analysis Modeling

- Focus on requirements, not technology.
- Use simple and clear models.
- Validate models with stakeholders regularly.

Example: Online Shopping System (Analysis View)

- Identify external entities such as user and payment gateway.
- Identify major processes such as order processing.
- Identify data stores such as product and order databases.

Benefits of Proper Modeling and Analysis

- Improved understanding of system requirements.
- Reduced rework and development cost.
- Better foundation for architectural decisions.